

Perfect Demand and Supply of $R_{\max} \leq 5$ Reserve Currencies

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Abstract

This paper develops a new theory of reserve-currency demand and supply under the hypothesis that the maximum number of realistic reserve currencies satisfies

$$R_{\max} \leq 5.$$

Rather than viewing currencies merely as media of exchange, stores of value, or units of account, we model reserve currencies as latent state variables generating a collection of observable monetary risk premia. The resulting framework unifies monetary economics, finance, political economy, geopolitics, international relations, welfare economics, warfare economics, statistics, machine learning, and representation learning.

A quintic reserve-currency architecture naturally emerges. We show that reserve-currency demand and supply may each be represented by quintic equilibrium equations and that a perfect-demand and perfect-supply condition collapses all admissible equilibrium branches into a unique monetary equilibrium point. This establishes a low-dimensional latent-factor interpretation of global reserve-currency systems and provides a mathematical foundation for a bounded reserve-currency architecture.

The paper ends with “The End”

Contents

1 Introduction

Modern monetary economics traditionally studies:

1. Currency production,
2. Currency circulation,
3. Currency management,
4. Currency withdrawal.

While these processes explain the mechanics of money, they do not explain why agents voluntarily hold currency.

A reserve currency competes with:

- Consumption goods,
- Strategic inventories,
- Precious metals,
- Sovereign debt,
- Equities,
- Real estate,
- Foreign currencies,

- Strategic military reserves.

The true problem of monetary economics is therefore:

What structural services does a reserve currency provide that induce economic agents to hold it?

This paper argues that reserve currencies generate a collection of observable risk premia arising from a lower-dimensional latent reserve-currency state.

2 The Five-Currency Hypothesis

We assume:

$$R_{\max} \leq 5.$$

Definition 1. *The reserve-currency state vector is*

$$c = (c_1, c_2, c_3, c_4, c_5).$$

The reserve-currency state vector may contain fewer than five active currencies, but never more than five.

Remark 1. *The hypothesis does not require exactly five reserve currencies. It merely imposes an upper bound.*

3 Reserve Currencies as Latent Factors

Traditional monetary models often write

$$M_d = f(Y, i, \pi).$$

where

- Y denotes income,
- i denotes interest rates,
- π denotes inflation.

We instead propose

$$M_d = h(z),$$

where

$$z = (c_1, c_2, c_3, c_4, c_5).$$

is a latent reserve-currency state.

Observable risk premia become projections:

$$p_i = g_i(z).$$

Examples include:

- Inflation premium,
- Liquidity premium,
- Safety premium,
- Convenience yield,
- Reserve premium,
- Geopolitical premium,
- Military-security premium.

4 Demand-Side Quintic Structure

Let

D_i = Demand for reserve currency i .

Define

$$F_D(x) = \sum_{i=1}^5 ((D_i - x)^5 - D_i^5).$$

Demand equilibrium is defined by

$$F_D(x) = 0.$$

Proposition 1. $x = 0$ is always a demand-equilibrium solution.

Proof. Substituting $x = 0$ yields

$$F_D(0) = \sum_{i=1}^5 (D_i^5 - D_i^5) = 0.$$

□

Expanding,

$$F_D(x) = -5xQ_D(x),$$

where $Q_D(x)$ is quartic.

Theorem 1. The demand equilibrium equation possesses at most five real solutions.

Proof. $F_D(x)$ is a quintic polynomial.

A quintic polynomial possesses at most five real roots.

□

5 Supply-Side Quintic Structure

Let

S_i = Supply of reserve currency i .

Define

$$F_S(y) = \sum_{i=1}^5 ((S_i - y)^5 - S_i^5).$$

Supply equilibrium is defined by

$$F_S(y) = 0.$$

Again,

$$F_S(y) = -5yQ_S(y).$$

where $Q_S(y)$ is quartic.

Theorem 2. The supply equilibrium equation possesses at most five real solutions.

Proof. The supply equation is a quintic polynomial.

Therefore it possesses at most five real roots.

□

6 Perfect Demand and Perfect Supply

Consider the system

$$\sum_{i=1}^5 ((D_i - x)^5 - D_i^5) = 0,$$

$$\sum_{i=1}^5 ((S_i - y)^5 - S_i^5) = 0,$$

and

$$x^2 + y^2 = 0.$$

Theorem 3 (Perfect Demand and Supply). *The system possesses exactly one real solution.*

Proof. Because

$$x^2 + y^2 = 0,$$

and

$$x^2 \geq 0, \quad y^2 \geq 0,$$

it follows that

$$x = 0, \quad y = 0.$$

Substituting into the demand equation yields

$$F_D(0) = 0.$$

Substituting into the supply equation yields

$$F_S(0) = 0.$$

Therefore

$$(x, y) = (0, 0)$$

is a solution.

No other real pair satisfies

$$x^2 + y^2 = 0.$$

Hence exactly one real solution exists.

□

7 Geometric Interpretation

Without the constraint

$$x^2 + y^2 = 0,$$

the demand equation may possess up to five real roots

$$x_1, \dots, x_5,$$

and the supply equation may possess up to five real roots

$$y_1, \dots, y_5.$$

The maximum number of equilibrium pairs would therefore be

$$5 \times 5 = 25.$$

The condition

$$x^2 + y^2 = 0$$

collapses the entire equilibrium lattice into

$$(0, 0).$$

Thus perfect demand and perfect supply eliminate all reserve-currency disequilibria simultaneously.

8 Economic Interpretation

The unique equilibrium

$$(x, y) = (0, 0)$$

corresponds to:

1. Zero aggregate reserve-currency demand imbalance,
2. Zero aggregate reserve-currency supply imbalance,
3. Simultaneous market clearing,
4. Global reserve-currency consistency.

In monetary terms, all reserve-currency deviations vanish simultaneously.

This is stronger than conventional market clearing.

It represents a complete collapse of all admissible reserve-currency disequilibrium states.

The End